

Dense b6 sweep shows a time-dependent tumor-control threshold

Using fixed model structure, finer b6 grid, 10,000-day ODE simulations

Main takeaway: day-365 control appears earlier than long-time control

Interpretation: the b6 response includes a finite-time delay effect plus a stronger long-time threshold

Model frozen

Direct CTL formulation
b5 fixed at $1e-4$
Only b6 varied

Finer b6 grid

40-point broad sweep
40-point threshold sweep
Threshold region: $4e-7$ to $9e-7$

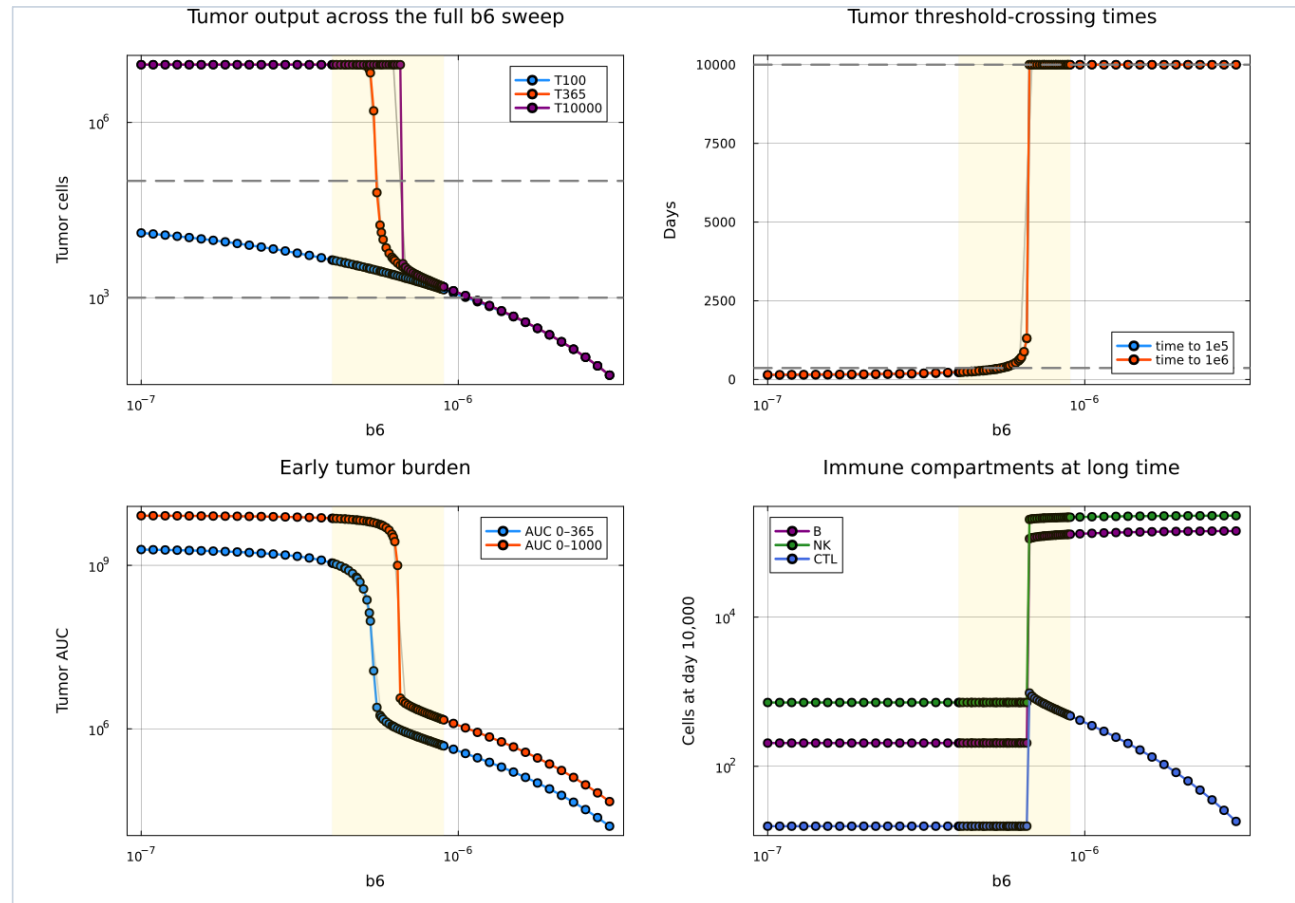
Long-time check

Simulated to 10,000 days
Compared T365 vs T10000
Recorded crossing times and AUC

Question for discussion: Is b6 acting like a true long-time transition, or a finite-time switch/delay?

Finding 1: The dense b6 sweep reveals a time-dependent threshold

Finer grade is represented in the dense summary and zoomed threshold plots



T365 drops sharply near the original threshold region.
T10000 shows that some apparently controlled cases eventually regrow.
Long-time control requires a larger b_6 than day-365 control.

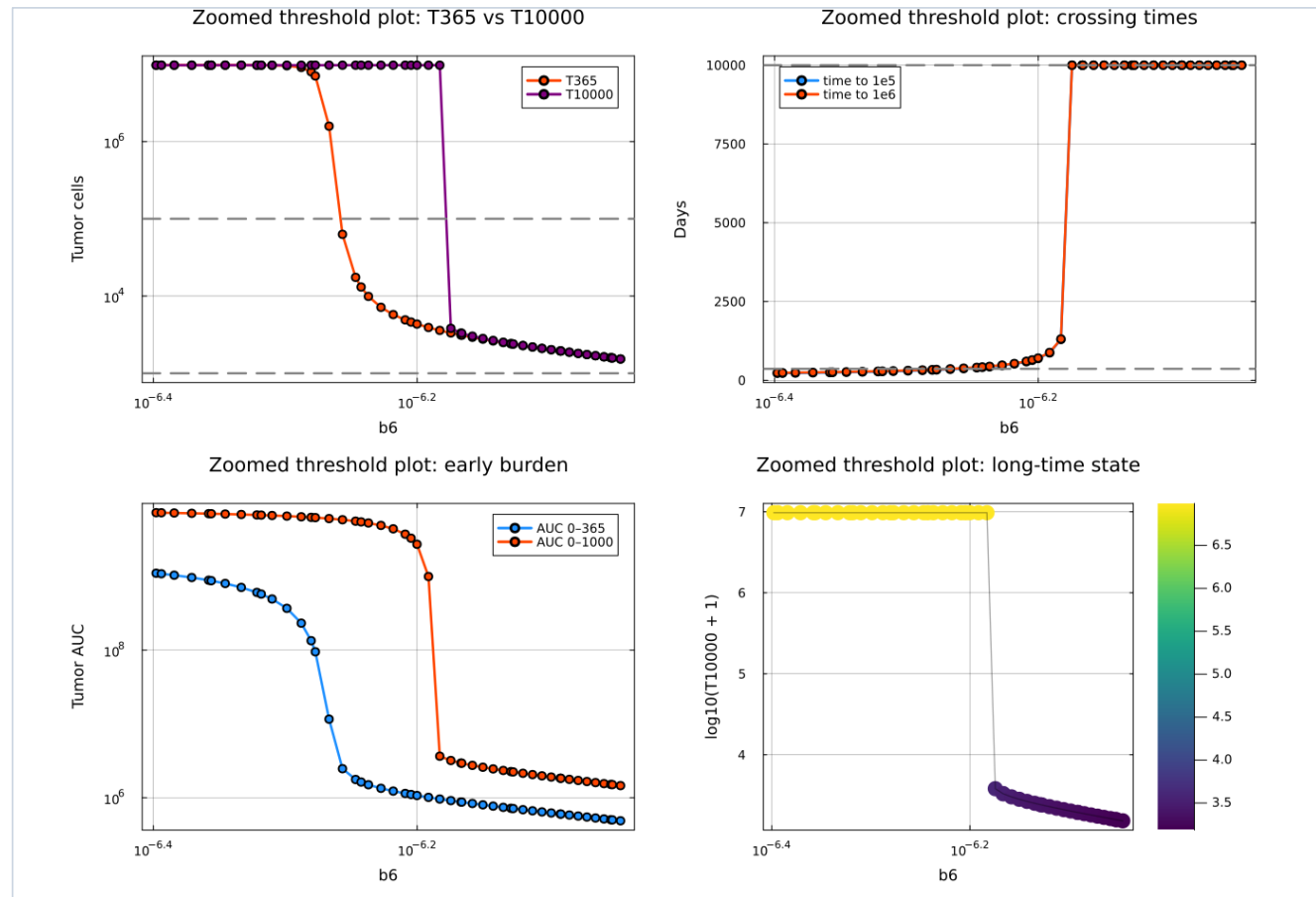
The 10,000-day endpoint changes the interpretation: not all day-365 suppression is permanent control.

Key values from the dense CSV:

First T365 < $1e5$: $b_6 \approx 5.54e-7$
First T10000 < $1e5$: $b_6 \approx 6.69e-7$
First T10000 < $1e3$: $b_6 \approx 1.15e-6$

Finding 2: Zooming in shows finite-time vs long-time tumor control

The finer b6 grid makes the shift between T365 and T10000 visible



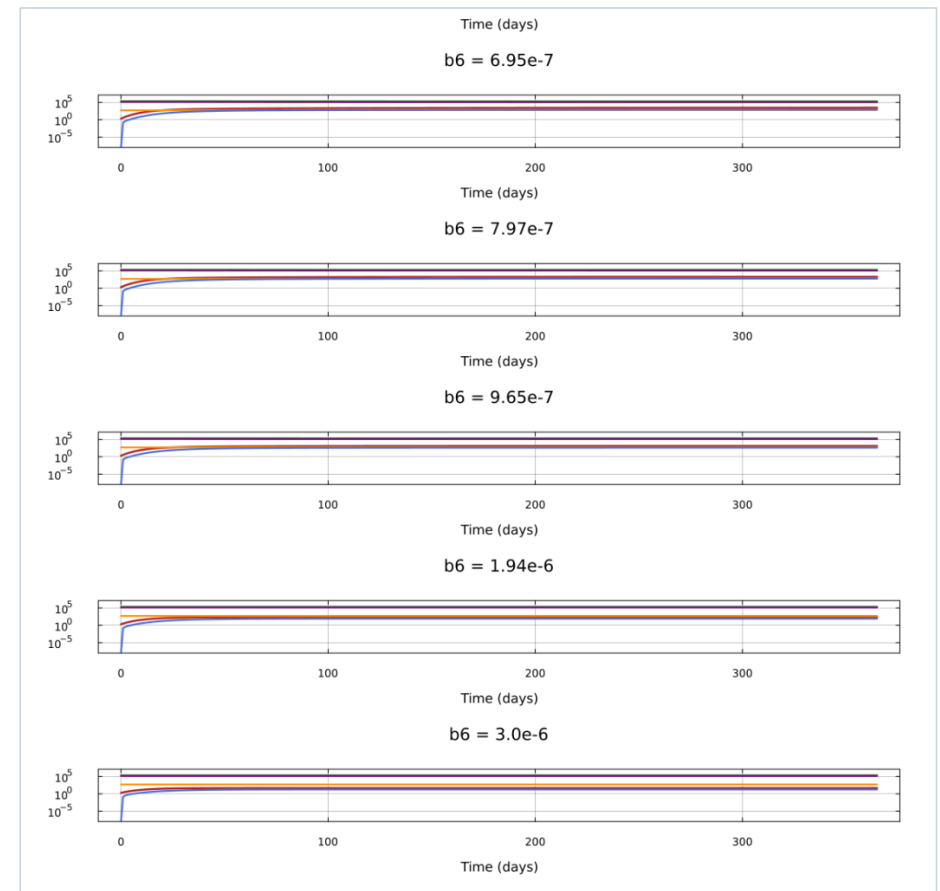
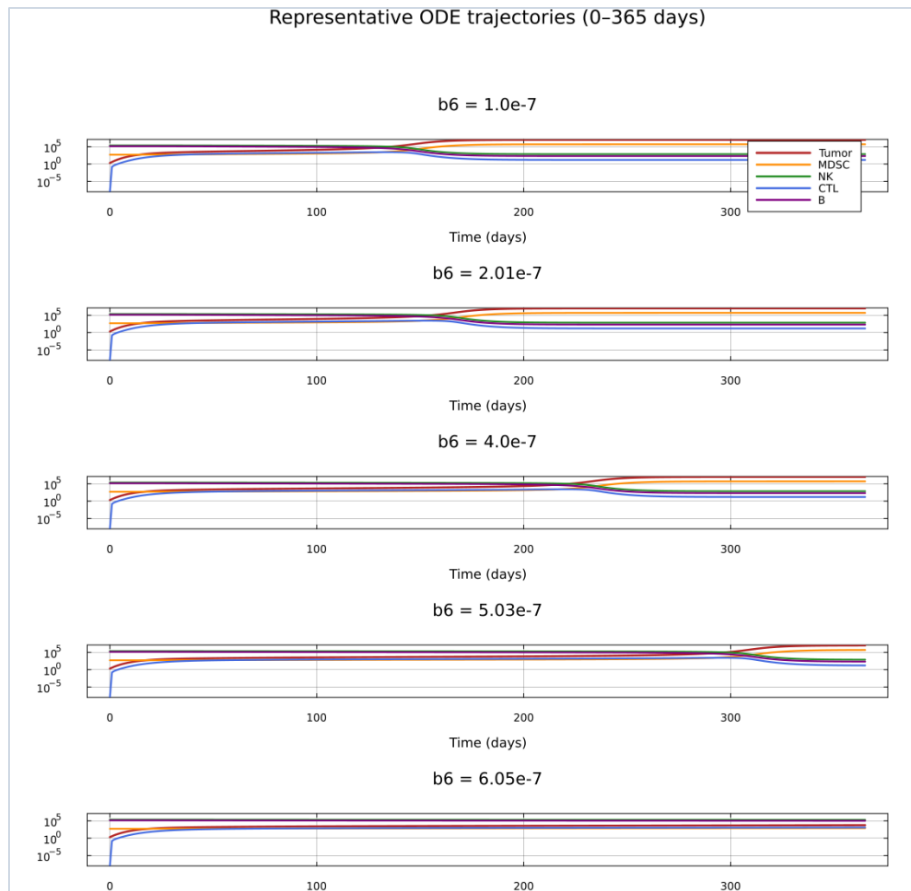
At day 365, tumor burden falls earlier in the $b6$ sweep.
At day 10,000, the transition shifts right: some low day-365 tumors regrow later.
Crossing times grow near the transition region, consistent with delayed growth/slow transition.

Interpretation: $b6$ creates both a finite-time suppression threshold and a stricter long-time control threshold.

$b6$:time-dependent, threshold-like

Finding 3: Representative 0-365 day trajectories show early b6-dependent suppression

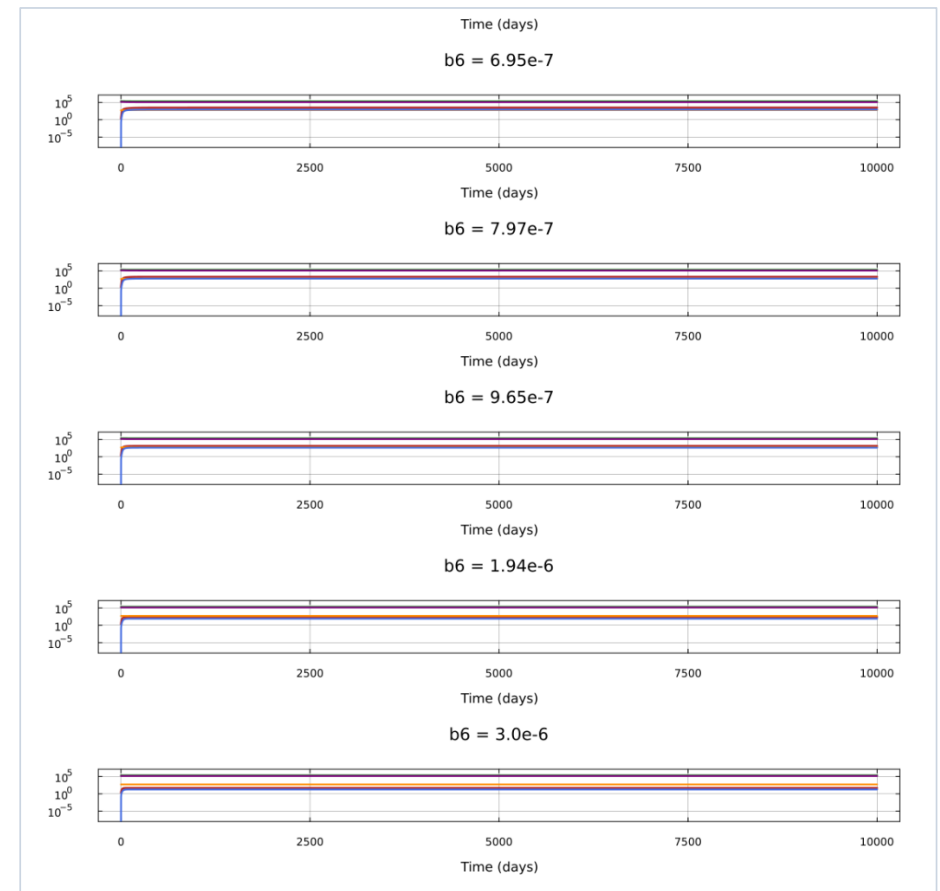
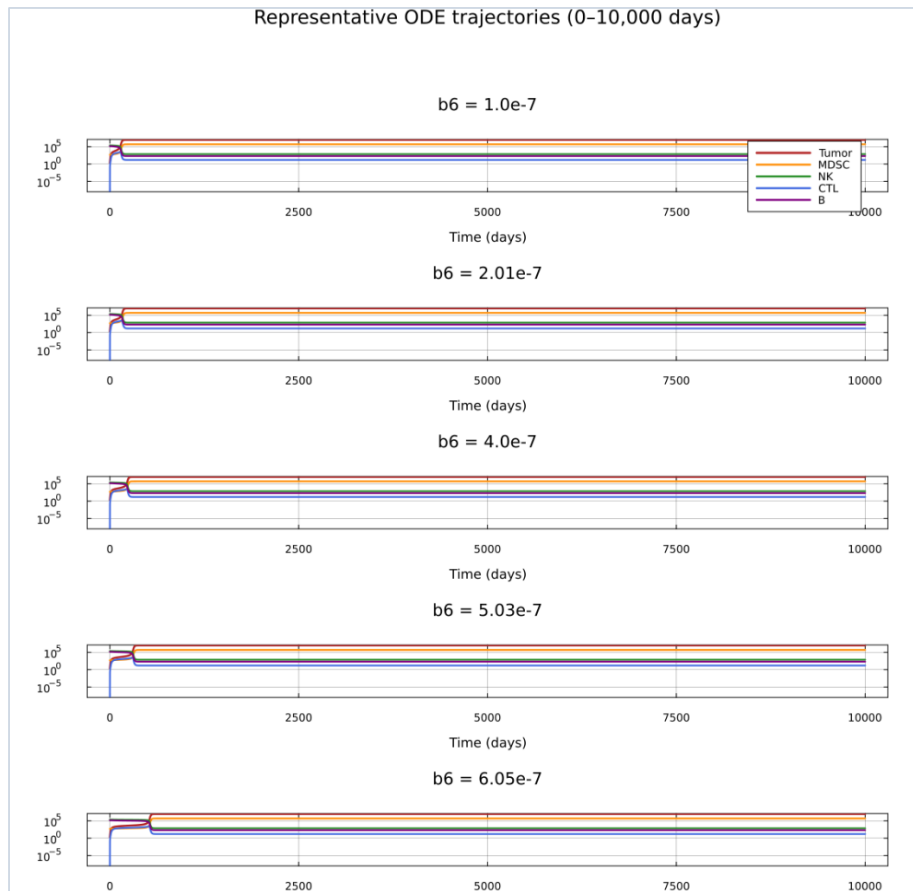
Short-time view preserves the early transient behavior



Early view: low b_6 remains tumor-dominant; near-threshold b_6 strongly delays tumor expansion.

Finding 4: 10,000-day trajectories separate transient delay from long-time control

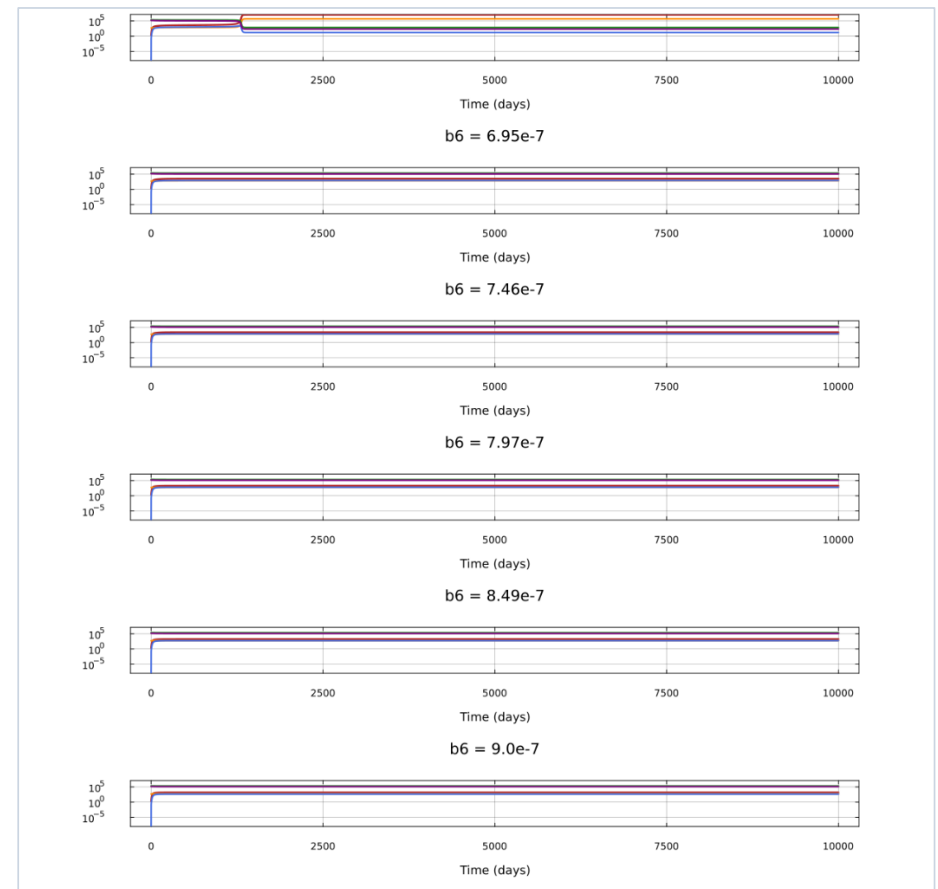
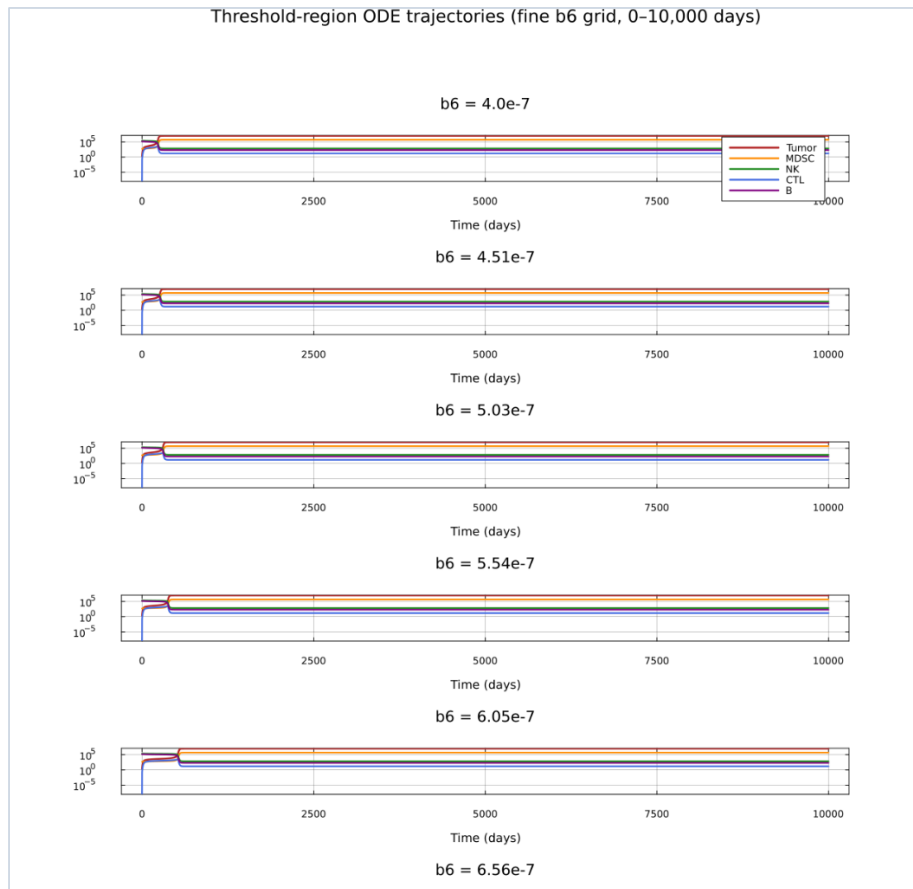
Long-time view checks whether day-365 suppression persists



Long-time view: some mid-threshold cases are delayed but not permanently controlled; higher b_6 remains low long-term.

Finding 5: Threshold-region trajectories resolve the fine-grid transition

Fine b_6 grid: $4e-7$ to $9e-7$, simulated to 10,000 days



This is the finer-grade resolves the transition zone instead of only sampling a few coarse b_6 values.

Summary: b6 is a strong B-specific parameter, but the threshold depends on time horizon

Current ODE interpretation

What is robust?

b6 clearly reduces tumor burden.
The response is threshold-like, not gradual across the full sweep.
ODE result is visible in trajectories and summary metrics.

What changed after 10,000 days?

Day-365 control appears earlier.
Some mid-b6 cases regrow by long time.
Long-time control requires a larger b6 cutoff.

Interpretation

The earlier threshold partly reflects finite-time delay.
A stricter long-time threshold is closer to $\sim 6.7\text{e-}7$ to $>1\text{e-}6$, depending on cutoff.
Next: test robustness to initial conditions.

Current conclusion: b6 produces a time-dependent threshold-like response; next analyses should test robustness and bifurcation-vs-switch behavior.

Next steps: test robustness and decide whether this is a switch or bifurcation-like transition

Proposed follow-up after dense b6 ODE sweep

1 Initial-condition robustness

Vary tumor, MDSC, NK, CTL, and B initial conditions one at a time.
Check whether b6 threshold persists or shifts.

2 Refined b5 x b6 phase diagram

Use longer time horizon (10,000 days).
Plot $\log_{10}(T_{10000} + 1)$ with a simple color scale and contour boundary.

3 Bifurcation-like vs sharp switch check

Run up-sweep and down-sweep using previous final states as next initial states.
Look for hysteresis or multiple attractors.

Questions for Prof. MacLean/Yingtong

Which tumor-control cutoff should we use: $T_{10000} < 1e3$, $< 1e4$, or another biological threshold?
Should the next priority be initial-condition robustness or the b5 x b6 phase diagram?